CDANet: Contextual Detail-Aware Network for High-Spatial-Resolution Remote-Sensing Imagery Shadow Detection

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***Abstract*— Shadow detection automatically marks shadow pixels in high-spatial-resolution (HSR) imagery with specific categories based on meaningful colorific features. Accurate shadow mapping is crucial in interpreting images and recovering radiometric information. Recent studies have demonstrated the superiority of deep learning in very-high-resolution satellite imagery shadow detection. Previous methods usually overlap convolutional layers but cause the loss of spatial information. In addition, the scale and shape of shadows vary, and the small and irregular shadows are challenging to detect. In addition, the unbalanced distribution of the foreground and the background causes the common binary cross-entropy loss function to be biased, which seriously affects model training. A contextual detail-aware network (CDANet), a novel framework for extracting accurate and complete shadows, is proposed for shadow detection to remedy these issues. In CDANet, a double branch module is embedded in the encoder–decoder structure to effectively alleviate low-level local information loss during convolution. The contextual semantic fusion connection with the residual dilation module is proposed to provide multiscale contextual information of diverse shadows. A hybrid loss function is designed to retain the detailed information of the tiny shadows, which per-pixel calculates the distribution of shadows and improves the robustness of the model. The performance of the proposed method is validated on two distinct shadow detection datasets, and the proposed CDANet reveals higher portability and robustness than other methods.**

***Index Terms*— Contextual information, deep learning, remote sensing, shadow detection.**

I. Introduction

SHADOWS are inevitable in remote sensing imagery; they appear when the direct beam is completely or partially obscured by an object [1], [2]. Shadow detection is an important prerequisite for shadow removal, and abundant works have been attempted to eliminate the negative effect of shadows on remote sensing images. However, shadow detection remains challenging, especially for high-spatial-resolution (HSR) images with abundant spectral and complex spatial patterns [3]–[8]. As shown in Fig. 1(a), shadows in HSR images are important prior knowledge for reconstructing 3-D information and determining the building boundary, whereas the continuity of gray values is disturbed by the wide range of shadows, which affect the interpretation of HSR images [9]–[11].

Many HSR image interpretation tasks, such as building identification, super-resolution reconstruction, and target tracking [12]–[15], require accurate shadow detection to minimize the impact of shadows [16]–[18]. For example, the vehicle is lost after entering the shadow region, as shown in Fig. 1(b). Accurate shadow area detection can provide auxiliary information for tracking the targets.

Various shadow detection methods have been proposed to reduce the impact of shadows, which can be summarized into geometric-based, property-based, and deep learning methods [19]–[21]. Geometric-based methods rely heavily on the prior position, which exerts a large effect on obtaining geometric information. Property-based methods circumvent this task, and the shadow dispersal can be directly obtained from the images. Among property-based approaches, the classical machine learning-based methods consider shadow detection as a task of information extraction by labeling shadow and nonshadow pixels in a handcrafted manner with basic visual features, such as texture, color, and illumination [22]–[24]. However, HSR images features contain rich spectral characteristics and diverse geometric structures. Thus, previous works that extracted shadows by handcrafted feature description required tremendous and time-consuming manual intervention and presented weak transferability.

Deep learning demonstrates superior inherent feature extraction capability in an end-to-end manner by employing a fully automatic feature simulation framework to learn essential information without prior knowledge [25]–[32]. This process solves the weaknesses of tedious processing steps in traditional shadow extraction methods and has been successfully applied to various remote sensing fields, such as hyperspectral image classification [33]–[37], object detection [38]–[41], and image reconstruction [42]. For natural images, convolutional neural networks (CNNs) have been successfully applied to shadow detection, which can be considered a binary segmentation task.

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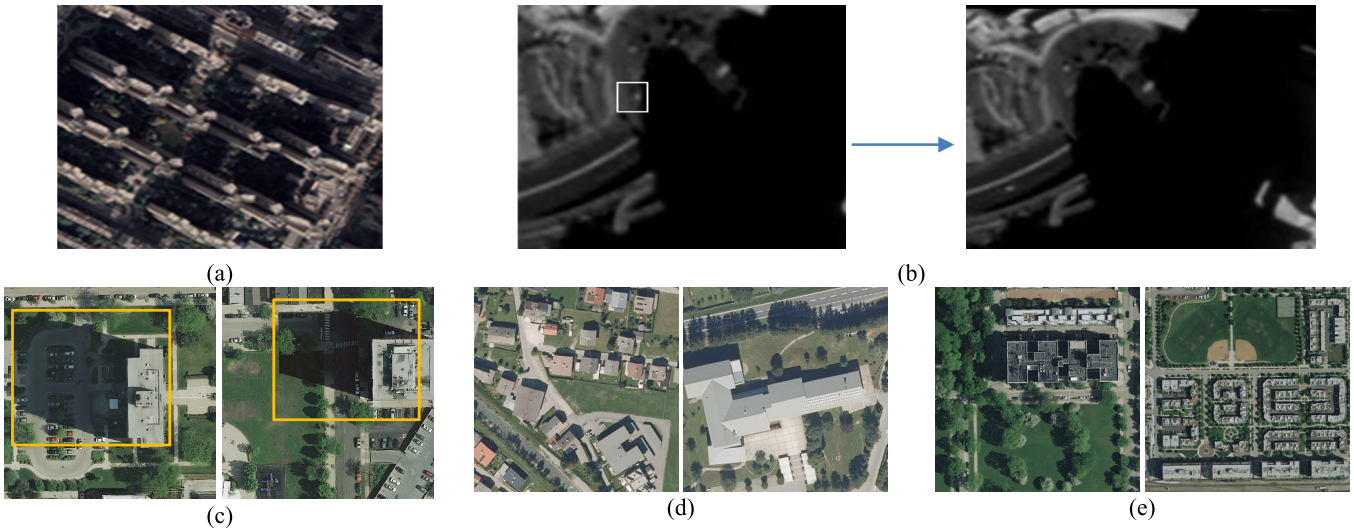


Fig. 1. Impact of shadows on HSR image tasks: (a) shadow affects the visual interpretation of the image; (b) target tracking may lose the target when the object enters the shadow region; (c) shadows of different buildings vary in size; (d) distribution of shadows in the observing scenes is messy; and (e) shadows formed by the trees are fragmented.

Among the CNNs, the classical fully convolutional networks (FCNs), U-shaped network (UNet), and deep fully convolutional neural network architecture for semantic pixel-wise segmentation (SegNet) have achieved impressive results for natural image shadow detection. Recently, a few deep learning-based methods have been introduced to HSR image shadow detection. In 2020, Jin *et al.* [43] combined a spatial-attention module and a novel loss function to improve the shadow detection effect. Luo *et al.* [1] then proposed the deeply supervised convolutional neural network (DSSDNet) as well as a newly constructed shadow detection dataset. In 2021, the edge-aware spatial pyramid fusion network (ESPFNet) [44] and shadow detection network based on multiscale spatial attention mechanism (MSASDNet) [45] were constructed to solve incomplete extraction of significant shadows based on aerial imagery dataset (AISD). Recently, Ufuktepe *et al.* [46] proposed a shadow mask estimation approach to provide shadow train data for deep learning-based shadow detection. However, previous deep learning-based methods remain difficult to deal with the complex and distinct shadows in HSR images, especially for small and irregular shadows displaying messy distribution, fragmented shapes, and varying scales.

The existing problems in previous works for HSR shadow detection mainly include the following.

1) For shadow detection, low-level local information can effectively assist in recovering features in the decoding process. However, due to the single convolution operation, the low-level information is often lost in the encoding process, which inhibits accurate shadow detection in HSR imagery.
2) During the encoding process, the shadow features get complicated and lack regularity. As shown in Fig. 1(c)–(e), the varying scale and messy distribution of the shadows are distinct characteristics, especially for HSR images. This aggravates the complexity and variety of the different types of shadows, which make it difficult to extract the global distribution and feature representation for detecting complete shadows.



Fig. 2. Shadows distribution in the dataset. Shadow and nonshadow samples are the regions marked with black and white. Shadow accounts for (a) 93% and (b) 5% of the entire image.

3) Shadow detection for HSR imagery can also be regarded as a semantic segmentation task. However, in the traditional deep learning-based shadow detection methods, the low-level local features and high-level global features cannot be effectively combined during the decoding process. This leads to the incomplete fusion of contextual information, and the critical information may be lost when restoring the images, which results in it being unrecognizable between the foreground and the background, such as the black objects and shadow areas.
4) As shown in Fig. 2(a) and (b), shadow accounts for 93% and 5% of the entire image, respectively. There was a frequent problem of foreground–background uneven cases for the shadows and the surroundings. Previous cross-entropy loss function-based shadow detection methods only concerned the shadow pixels and ignore other nonshadow pixels. It led to the cross-entropy loss results in less information learned from the model because of the limitations of available features. In addition, due to the inadequate detail perception, the cross-entropy loss function has a weak detection ability for tiny shadows.

To solve the above-discussed problem, the contextual detail-aware network (CDANet) for HSR imagery shadow detection is proposed. In CDANet, the discriminative feature representation is generated by the double branch (DB) block, which produces two outputs: one carrying low-level local information for decoding and another for skip connection. The residual dilation (RD) convolution is used for the

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high-level semantic information to mine potential global feature distribution and details. Then, adjacent features are combined in context semantic fusion connection (CSFC) as auxiliary information during the decoding process, to predict salient shadow regions. In addition, a hybrid loss function combining binary cross-entropy (BCE) loss and Lovasz hinge loss is proposed for the foreground-background uneven shadow scenarios, and both regular shadows of buildings and tiny shadows of the tree are effectively captured.

The main contributions of this article are as follows.

1) The CDANet framework is proposed to address the difficulties of HSR imagery shadow detection. CDANet is constructed with a DB module-based encoder, CSFC in the decoder, and the hybrid loss function. By using different convolution kernels to capture the shadow information, the DB block can effectively focus on different scales of information at the same resolution. Based on the DB block, the local details that can increase the shadow weight during encoding are recovered, especially for the tiny and fragmented shadows in HSR images.
2) CSFC with an RD module is proposed to effectively fuse the associated local and global shadow information in HSR imagery. The RD module is introduced at the beginning of CSFC to efficiently compress and extract global high-level shadow distribution and variations. To improve the discriminative ability of the shadows that are similar to the surroundings, the high-level shadow feature extracted by RD and low-level local details from DB are combined in CSFC as contextual shadow semantics. The integration of shallow and deep features provides a multilevel feature description to highlight prominent shadows and weaken nonshadowed regions.
3) The hybrid loss function is designed to be flexibly applied to the HSR scenes with mostly irregular and fragmented shadows, which results in a very small proportion of shadows compared to the background. This novel loss function consists of two parts: one where messy shadows are focused by the Lovasz hinge function and another where regular shadows are noticed on the BCE loss. Based on the hybrid loss function, the problems of uneven foreground–background and incomplete detection of tiny shadows are solved.

The proposed CDANet framework was evaluated on two publicly published HSR shadow detection datasets, i.e., the AISD [1] and the shadow semantic annotation dataset (SSAD) [47]. Experimental results on the two distinct datasets confirm the effectiveness of CDANet.

The remaining of this article is organized as follows. The related research is discussed in Section II. In Section III, the proposed CDANet framework for shadow detection is explained in detail. In Section IV, the description of the dataset and the experimental results are presented. Finally, the conclusions are drawn in Section V.

II. Related Works

The CDANet framework is constructed on the encoder–decoder structure. In this section, we briefly introduce the current status of shadow detection and the classical encoder–decoder structure.

A. Current Status of Shadow Detection

Shadows have always been the object concerned in remote sensing because of their interruption of other missions. These methods are divided into three main categories according to the properties of shadows: geometrical methods, property-based methods, and deep learning methods. Geometrical methods determine the location of shadows through modeling the geometrical optics based on the surface altitude and the sensor position information. Li *et al.* [48] simulated geodetic shadows from a digital surface model and the sun attitude. Property-based methods have been broadly employed in the last few years because of their simplicity and availability. In detail, property-based methods can be divided into machine learning-based methods and thresholding-based methods. Machine learning-based methods usually deem shadow detection as a classification task. Perceptron classifiers [49], support vector machines [50], [51], and other typical learning-based classifiers are employed to solve this task. Machine learning-based methods could achieve good performance for images, but these methods may crash when the scene is complicated. Most dark objects are often recognized as shadows when the classifiers are applied to HSR images. Thresholding-based methods usually use different ways to convert the image into a specific shaded graph and then identify the final threshold by Otsu or bimodal histogram [52], [53]. Hong *et al.* [36] proposed an improved shadow index (ISI) combining color space with a near-infrared response (NIR) band for shadow detection. The ISI was reconstructed by the objects acquired from the mean-shift algorithm to weaken the noise interference and improve integrity. Finally, threshold segmentation was applied to obtain the shadow mask. Thresholding-based methods identify each pixel separately and barely consider the spatial correlation between the adjacent pixels. Thus, the sunlit dark objects and shadowed bright objects are hard to distinguish. Shadows are characterized by low brightness, high saturation, and large color dispatch. Thus, color-transformation methods motivated by thresholding-based methods are widely used. Shadows are extracted by some indicators after color transformation. For example, Sarabandi *et al.* [54] converted imagery based on color-invariant indicators and proposed a novel method. Although these methods exclude the most obvious nonshadow regions, they usually require additional steps to eliminate partial recognition errors, such as the salt-and-pepper phenomenon. Meanwhile, some methods combine 3-D data and sunlight models to detect shadows [48], [49]. However, their methods are limited when 3-D information is unavailable.

With the tremendous advancement of artificial intelligence in previous decades, many deep learning models have been successfully applied. The shadow detection task can be considered a subpart of segmentation. Many semantic segmentation approaches, such as FCN, UNet, SegNet, pyramid scene parsing network (PSPNet), and DeepLabv3+ [55]–[59], can be well migrated to the shadow detection task, and they yield better results than traditional methods. Based on the semantic segmentation approach, many shadow detection networks have

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been proposed comprehensively. In the natural shadow detection field, Le and Samaras [60] and Wang *et al.* [61], [62] used a new perspective on shading as an instance object and predicted self-shadow and cast shading. Nguyen *et al.* [63] first applied generative adversarial network (GAN) networks to the shading problem and developed a novel conditional GAN architecture with a tunable sensitivity parameter. Hu *et al.* [64], [65] and Zhu *et al.* [66] developed the direction-aware spatial context module and recurrent attention residual module to enhance the spatial contextual information during shadow detection. Chaki [67] used generated fuzzy instances to interfere with system training to make robust from the change in brightness and contrast. Chen *et al.* [68] proposed the first deep-learning-based method for video shadow detection [triple cooperative video shadow detection network (TVSD-Net)], which uses a three-layer parallel network cooperatively to learn discriminative representations within and between videos. Chen *et al.* [69] proposed a semisupervised multitask baseline model [multitask mean teacher (MTMT)] to extract shadow region, shadow edge, and shadow count information separately. The extracted information is assigned to the student and teacher networks and it encourages both networks to be consistent in their predictions of the information. Therefore, semisupervised shadow detection can be effectively achieved by using a supervised network that is consistent with an unsupervised network.

In recent years, many scholars have introduced deep learning to the field of remote sensing for extracting complex shadows in HSR images. Luo *et al.* [1] proposed a deep learning model (DSSDNet) applied to aerial imagery, which uses an encoder–decoder residual structure and a deep supervised fusion module. Jin *et al.* [43] combined a spatial-attention module and a novel loss function to improve the training effect. Zhang *et al.* [47] proposed a recurrent shadow attention model to retrieve fine-scale urban land model classes in shadows. Given that the primary purpose of this model is to increase the classification accuracy of land, its shadow detection accuracy is general. Luo *et al.* [44] combined a parallel space pyramid (PSP) structure, an edge detection module (EDM), and an edge-aware multibranch integration (EMI) to construct an ESPFNet for remote sensing salient shadow detection. Liu *et al.* [45], based on a multiscale spatial attention mechanism, proposed MSASDNet to detect aerial remote sensing shadows. Ufuktepe *et al.* [46] generated a shadow mask by 3-D point clouds combined with known solar angles. The novel shadow mask self-supervision approach develops new deep architectures for shadow detection. However, the above-discussed methods do not consider sufficient contextual semantic information when encoding and decoding. The shadows in HSR images often display complex and challenging characteristics that were ignored by previous works. In addition, these networks often lack multiscale information of different layers, making the varying scale and messy distribution of HSR shadows difficult to detect.

B. Encoder–Decoder Structure

In recent years, many types of encoder–decoder structures have been proposed for different task requirements, including semantic segmentation [55]–[59], object detection [38], [40], and machine translation [70], [71]. In general, the encoder–decoder structure includes two parts. The encoder compresses the input information, captures the high-level feature map, and reduces the dimensions of the feature map. Then, the decoder recovers the high-level feature map to the same dimensions as the input image.

Models based on the encoder–decoder structure have shown significant performance on semantic segmentation. For example, SegNet converts max-pooling in the decoder to improve resolution [58]. RefineNet uses residual connections to minimize memory usage and enhances feature fusion between modules [72]. PSPNet, DeepLabv3+ [59], and other encoder–decoder networks have emerged successively. Although these networks have improved classification accuracy, the complexity has also increased and is limited to be migrated to remote sensing fields.

Shadow detection is a semantic segmentation task, and U-Net is simpler than other encoder–decoder networks with good portability in semantic segmentation tasks. Hence, the CDANet framework uses a symmetrical encoder–decoder structure similar to U-Net, where the features with the same size at different levels in the encoder and the decoder could hierarchically be aggregated by skip connection. When U-Net is directly applied to HSR shadow detection, they do not consider the fragmentation and uneven distribution of shadows. To capture the distinct characteristics of shadows in HSR imagery, the CDANet framework further improves U-Net by designing with the DB module, RD module, CSFC, and hybrid loss function to solve these problems.

III. CDANet for HSR Images

The proposed CDANet is designed to improve the accuracy of shadow detection. It is divided into three main parts. First, a DB module is used in the encoder to alleviate the information loss problem. The encoder is enhanced by incorporating the DB module. Continuously, with the increase in model complexity, the RD module can effectively suppress gradient explosion and disappearance. Global and local information can also be integrated by the CSFC. This operation can convey prominent semantic information based on increasing feature hierarchy. Finally, the hybrid loss function is applied to generate comprehensive shadow segmentation maps. The overall architecture of CDANet is shown in Fig. 3.

A. Encoder With DB Module

Spatial information is momentous in shadow extraction. However, some spatial information is lost because of the frequent convolution operation in deep learning. Inspired by the above-mentioned facts, this article uses a DB to alleviate information loss in encoding [73]. As a result, the loss of spatial information is recovered. The DB module has two essential impacts. It helps the network retain shadow features and maintain those contextual relations. Then, it is beneficial for gradient descent through the numerous iterations of the model.

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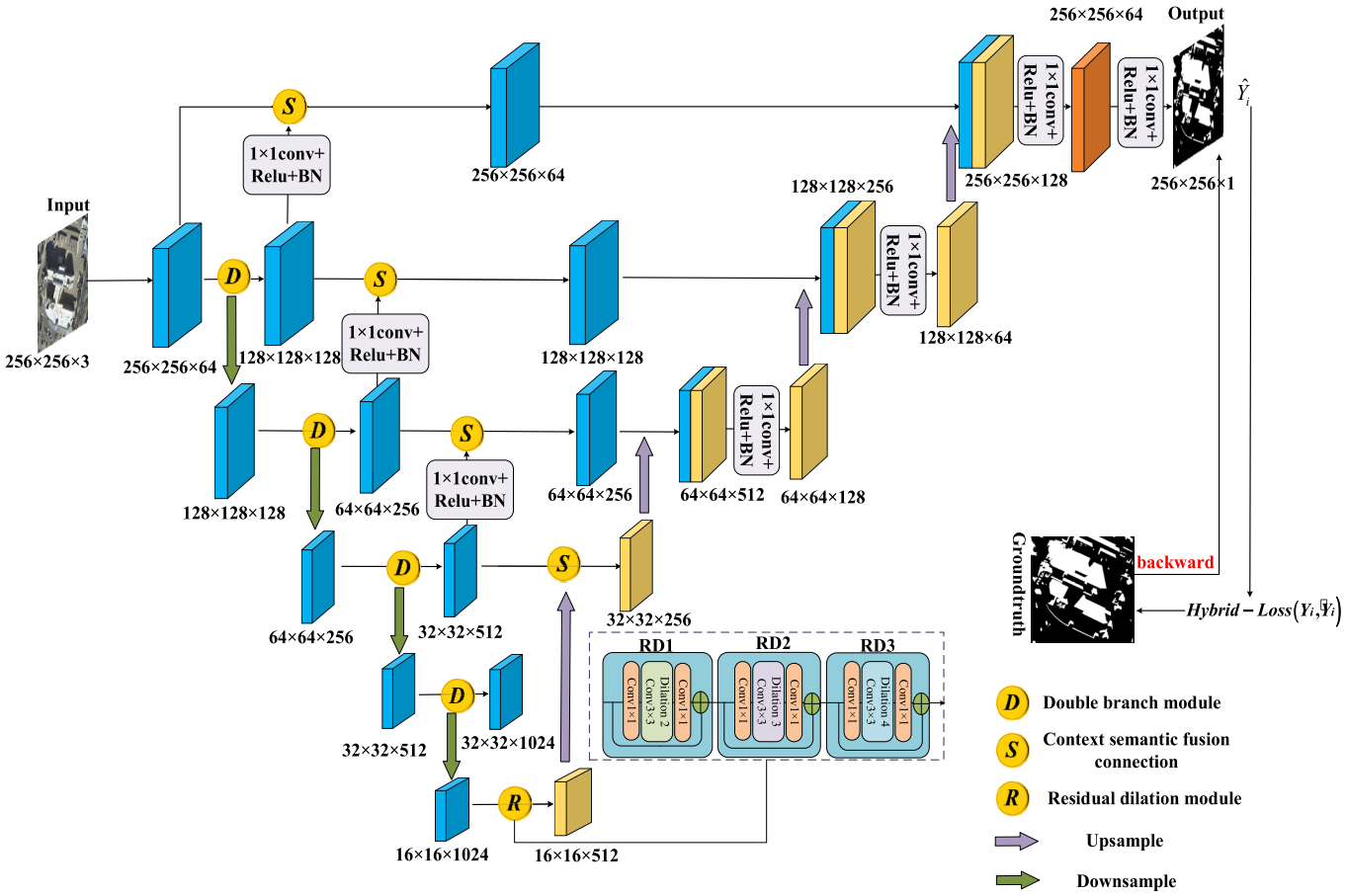


Fig. 3. Shadow detection based on the CDANet framework for HSR images.

The size of the convolution kernel not only controls the theoretical receptive field [74], but also affects the weight of information in the effective receptive field region. The effective receptive field is only a part of the theoretical receptive field. As the effective receptive field is Gaussian distributed, the shadow information distributed in the edge may not be fully captured. When using the same size-based convolutions, the extracted critical features for shadows are usually redundant [75]. Therefore, the DB module is designed to utilize different sizes of convolutional kernels for discovering distinct shadow information at multiple scales. Fig. 4 displays the components of the DB module. The left part is a process of information extraction. It applies a double layer with a kernel size of three sliding in steps of 1 to extract features from input. In this process, we use the padding of 1 for filling images to ensure the feature size is consistent with the input. The right part makes the input features pass through a single layer of 1×1 convolution block and the other reserve the input features without any convolution operation. It can be regarded as a process to mitigate shadow information loss. Then, the features extracted from this part are combined with those from the left branch to effectively complement the feature information by increasing the weight of the true shaded region and weakening the nonshaded region.

$O1$ represents the features that need skip connection, and $O2$ represents the feature information used for transmitting down. As shown in Fig. 3, the DB module is applied to each encoding used to reserve the shadow information.

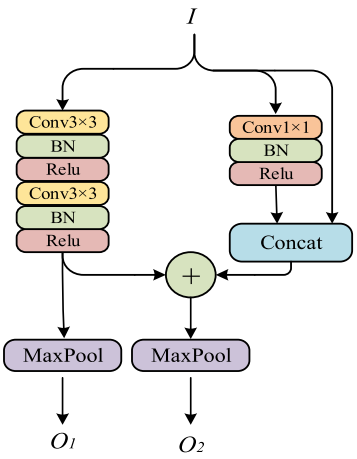


Fig. 4. Structure of the DB module.

B. CSFC With RD Module

In CNN, the low-level shadow features are too messy, and the deep shadow features usually lack regularity. Although detailed information can be extracted using the DB module,

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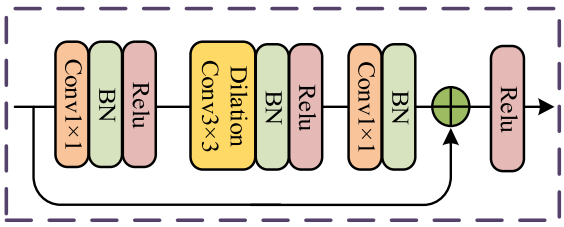


Fig. 5. Structure of the RD block.

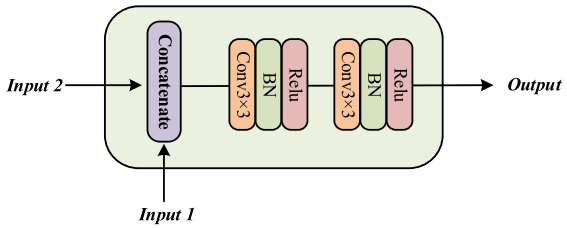


Fig. 6. Structure of the CSFC.

the global spatial relationship and scale variations of shadow objects remain to be complicated to be effectively captured. To extract the global high-level features from the complex relationship, CSFC with RD is proposed to effectively fuse the shading contextual features at different levels.

The RD module at the beginning of CSFC can effectively suppress the disappearance or explosions of gradients [66]. The residual network also can solve this phenomenon by adopting shortcut connection [76], but it will generate a large algorithm complexity when directly applied in the high complexity and large scale of HSR imagery. The RD block uses dilation convolution instead of traditional convolution. It can explore the relationship between more distant pixels in shadow features without increasing the number of parameters. The convolution with a kernel of 1 changes the channel for concreting operation to make the final feature size consistent. The output of an RD module is obtained by

$$(O) = \mathrm{RD}_3(\mathrm{RD}_2(\mathrm{RD}_1(I) + I) + \mathrm{RD}_1) + \mathrm{RD}_2 \tag{1}$$

where RD_1, RD_2, RD_3, I, and O represent the RD block, the input shadow feature (from the previous downsampling), and generated outputs (O). RD_1 and RD_2 do not change the channels and size of the input, whereas RD_3 changes channels from 1024 to 512 for matching with the previous low-level shadow feature.

The RD_1, RD_2, RD_3 blocks have the same structure, as shown in Fig. 5. The only difference lies in the different sizes of the dilated convolution. Specifically, the RD block begins with a 1×1 convolution, which is followed by a dilated convolution. Finally, a 1×1 convolution is used for resizing the feature maps, and it is connected with the input feature map. In particular, the size of the dilation convolution is different among each RD block, with 2 for RD_1, 3 for RD_2, and 4 for RD_3, respectively. The RD block mines the global high-level shadow distribution and variations that contribute to the refinement of the local shadow features. The cascade structure-based RD module is shown in Fig. 3.

The CSFC is designed to combine two adjacent layers to overcome the adverse influence of confusing spatial information and learn robust and typical information. As a result, it can produce neat and salient information in a coarse-to-fine manner.

The structure of the CSFC is shown in Fig. 6. Each fusion requires the feature through a deconvolution and pooling operation. It makes two neighboring shadow feature maps have the same size. Through the concatenation operation, the corresponding feature map makes a skip connection in the decoding process. By adding an auxiliary feature enhancement method to the intermediate steps, each upsampling can learn from the neighboring levels directly, leading to a highly flexible shadow feature fusion scheme.

C. Error Calculation Based on Hybrid Loss Functions

In deep learning networks, the loss function is a vital component. It can calculate the gap from the model predictions to the ground truth. Finally, through the gap to guide the model into the optimal state. BCE loss is commonly used for binary segmentation tasks, which is denoted as

$$L_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=0}^{N}\left(y_i \log \sigma\left(y_i'\right) + (1 - y_i)\log\left(1 - \sigma\left(y_i'\right)\right)\right) \tag{2}$$

where $y_i \in \{0, 1\}$ expresses the ith pixel of ground truth, and the corresponding network predicted results are represented by $y_i' \in \{0, 1\}$.

For each image, the proportion of the shadow area is unstable, and the shape of the labels is more complex and disorderly. If the original BCE loss is directly used to supervise the network, the imbalanced distribution of class could cause inadequate learning ability of the network. We replace BCE loss with hybrid loss to avoid the negative consequences, which combine Jaccard loss with Lovasz hinge and BCE loss. The new loss function is flexible for the prediction results and ground truth at each iteration.

Jaccard loss is a pixel-by-pixel calculation and does not suffer from the effects brought by uneven foreground–background distribution. While the Jaccard index is usually calculated for each pixel on the entire dataset, calculating the Jaccard index on a single image can remove the bias from the large categories. Thus, we use a loss function that combines Lovasz hinge and Jaccard loss to solve the segmentation task of two-valued images. For ground truth y^* and predicted output $\tilde{y}$, the incorrect prediction set pixels for class c is defined as

$$M_c(y^*, \tilde{y}) = \left\{y^* = c, \tilde{y} \neq c\right\} \cup \left\{y^* \neq c, \tilde{y} = c\right\} \tag{3}$$

where y^* represents the ground truth, and the Jaccard loss can be written as a function of the incorrect prediction set

$$\Delta J_C : \{0, 1\}^p \rightarrow \frac{|M_C|}{|\{y^* = c\} \cup M_C|} \tag{4}$$

$$\tilde{y}_i = \mathrm{sign}(F_i(x)) \tag{5}$$

$$m_i = \max\left(1 - F_i(x) y_i^*, 0\right) \tag{6}$$

$$\mathrm{Loss}(F) = \bar{\Delta}_{J1}(m(F)). \tag{7}$$

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We recognize pixel subsets with their indicator vector in the discrete hypercube $\{0, 1\}^p$ for convenience of notation. First, L_{lovasz} uses Jaccard loss to optimize the foreground class. Then, for each image x, we have y_i^* as the ith pixel of ground truth, $F_i(x)$ as the value of the ith output scoring function, and $\tilde{y}_i$ as the predicted label. Thus, the hinge loss for a predicted pixel i is m_i. Under this scenario, the vector of hinge losses $m \in \mathrm{R}^+$ is the vector of errors discussed before, and $\bar{\Delta}_{J1}$ is a Lovasz expansion for Δ_{J1}. Its specific push-to equation is in [77]. Loss(F) is the Jaccard loss in combination with Lovasz hinge.

Two loss functions are added to form a new loss function for compensating the deficiency caused by cross-entropy loss, and the final loss is calculated by

$$L = L_{\text{BCE}} + \lambda L_{\text{Jaccard-hinge}}. \tag{8}$$

In addition, the different weights of the two loss functions bring different effects. For example, too much weight of the BCE loss leads to the loss of tiny shading information. On the contrary, too much weight of the Jaccard loss with Lovasz hinge leads to the missing of regular shading. To obtain a stable loss function, we add λ as a coefficient and determine the optimal loss function weights by experiment. We will describe the detection effect of different λ values in Section IV-G.

IV. Experiments and Analysis

The proposed CDANet is compared with other shadow detection methods, such as DSSDNet [1], bidirectional feature pyramid network with a repeated attention residual module (BDRAR) [66], and MTMT module [69]. Furthermore, shadow detection can be considered as a semantic segmentation task, such as road extraction and building extraction, and three typical semantic segmentation networks (i.e., FCNs [55], U-Net [56], and DeepLabv3+ [78]) are selected for comparison.

A. Experimental Dataset

Each dataset has three main components. A training set occupies most parts of the dataset. It is used to fit the model and train the model. When several models are trained from the training set, to identify the best model, the validation set is employed to predict the validation set data using each model and record the accuracy of the models. Finally, the testing set is used to determine the capability of the optimal model. Images are selected from a novel AISD collected and labeled by Luo *et al.* [1]. The dataset is designed especially for shadow detection, and it has the characteristics of high resolution and multiple land-cover classes. Five regions of the world are covered in this dataset, namely, Tyrol, Vienna, Austin, Chicago, and Innsbruck (like Fig. 7). The training set includes 412 images and related masks, whereas the test set and validation set contain 51 images and related masks, respectively. SSAD has a total of 103 image patches. The size of each image patch is 500 × 500 pixels. The scenes in this dataset are very complex. Therefore, manual labeling alone is not practical. Zhang *et al.* [47] initially used a natural image dataset for migration learning to generate a predetermined sample block and then manually labeled based on the sample block. The shaded labels are accurate and comprehensive when using this method.

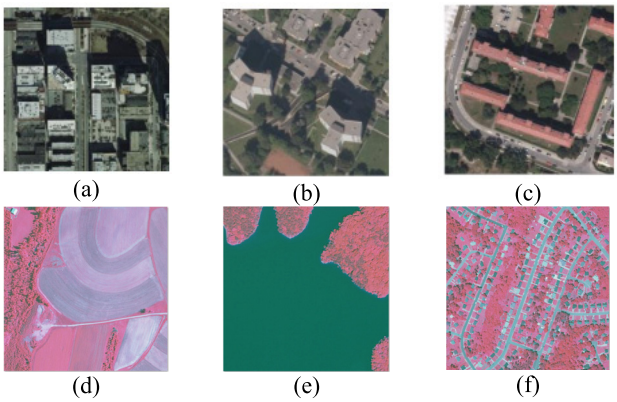


Fig. 7. (a) Austin; (b) Tyrol; and (c) Vienna in AISD images dataset and (d)–(f) in SSAD images dataset.

Fig. 7 shows a representative sample of each region in the AISD and SSAD. As shown in Fig. 7(a), Austin and Chicago have dense high-rise buildings, which shadows produced are more obvious. As shown in Fig. 7(b), the trees are widely distributed, and the shadow presents a fragmented distribution in Tyrol. Finally, as shown in Fig. 7(c), the architecture comparison in Vienna combines regular buildings and irregular trees. We test our model in the SSAD to demonstrate the generalizability of the model. The shadow in this dataset is obvious. However, given that water reflection is similar to the shadow, the water also appears as black as the shadow, which makes shadow detection difficult.

Much training data can effectively prevent overfitting, and the model can effectively converge by a large batch size. However, using an excessively large single image size limits the use of a large batch size. Individual images are cropped to produce small patches. We refer to [1] and set the patches to be 256 × 256 with a stride of 64. We also randomly add more complex images, which is a data enhancement form to increase complexity and validity. By the above-mentioned data augmentation, for the AISD, 13 742 image patches are taken for the model training. For the SSAD, 79 images are selected as a training set and cropped according to the same stride with the AISD, thus providing 1900 patches for training. Except for cropping the data, redundant data augmentation to enhance the training process of the model is not performed to explore the true validity of the model.

B. Experimental Setting

The Adam optimizer is used to train the model with an initial learning rate of 0.0001 and L2 weight regularization with 0.0005. We use a learning rate scheduler to reduce the learning rate by 0.1 times the original learning rate when the monitoring volume stops descending more than twice. The network is trained 100 epochs for the AISD and SSAD. We uniformly select the best-performing epoch on the

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evaluation set for testing. To compare the robustness of the model, we train each model ten times and then average it to find the confidence interval of the model. All experiments are conducted with an NVIDIA GeForce GTX 2080 Ti GPU. A batch size of 10 is used in every experiment. A hybrid loss is employed to guide the entire network training. The cropped dataset is used for training and evaluation. Accuracy evaluation on both AISD and SSAD is executed.

The complex structure of our network causes the inefficiency of backward gradient transfer, which can easily lead to the overfitting of posterior neurons. Meanwhile, the neurons of the forepart are undertrained, leading to overfitting. Therefore, we use dropout to randomly freeze some of the neurons in the encoder, which makes the gradient backward transfer efficient. All neurons need to be activated during testing. Thus, the frozen parameters must defrost.

To make the comparison fair, all deep learning models are retrained with the same training dataset, employed with BCE loss and the same hyperparameter setting. Then, an ablation study is conducted to verify the effectiveness of CDANet.

C. Evaluation Indexes

To analyze the performance of different methods quantitatively, we choose the F1-score (F1) as an evaluation criterion in the subsequent accuracy assessment. The metrics are expressed as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\% \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\% \tag{10}$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100\%. \tag{11}$$

In (9) and (10), true negative (TN) represents the quantity of true nonshadow pixels that are correctly identified, true positive (TP) is the total quantity of true shadow pixels that are correctly identified, and false negative (FN) is the number of true shadow pixels that are detected as nonshadow pixels. False positive (FP) is the quantity of nonshadow pixels that are mistakenly identified as shadow pixels. Precision focuses on the proportion of data that TP samples are predicted to be positive (and may contain negative samples). Recall focuses on the proportion of data in the TP samples (which does not include any negative samples), which are correctly predicted. F1 considers both precision and recall in our binary classification task, and the performance of the model is better with a higher F1 index.

D. Experiment 1: Novel AISD

The proposed CDANet is compared with FCN, U-Net, BDRAR, DSSDNet, DeepLabv3+, and MTMT on the AISD. The average shadow detection accuracy for the testing set is shown in Table I. The proposed CDANet achieves the best performance and robustness, with an F1 score of $92.99\% \pm 0.17\%$. Compared with those of FCN, U-Net, BDRAR, DSSDNet, DeepLabv3+, and MTMT, the average F1 score of CDANet achieves a significant improvement of 5.88%, 4.53%, 4.18%, 1.27%, 4.72%, and 2.31%, respectively.

TABLE I
COMPARISON OF DIFFERENT SHADOW DETECTION METHODS ON AISD

Model Name	F1(%)
FCN	87.11±0.33
U-Net	88.46±0.50
BDRAR	88.81±0.93
DSSDNet	91.72±0.32
DeepLabv3+	88.27±1.58
MTMT	90.68±0.29
Ours	**92.99±0.17**

The visualization results are shown in Fig. 8. By comparison, we can see some differences in detail. As shown in Fig. 8(c), FCN has a small amount of noise in shadows, and the edges of the shadows are also examined incompletely. The main reason is FCN does not consider the relationship between pixels and loses a large amount of spatial information. The results of the U-Net method are shown in Fig. 8(d), where some light-shaded regions are incorrectly identified as nonshadow regions. The reason is that the skip connection of U-Net is made only on the same specification of feature maps. Thus, it lacks multiscale information features as a compliment. BDRAR is mainly used in simple natural images. When it is used to detect complex scenes, as shown in Fig. 8(d), although it can detect salient shadow areas, the shadow boundaries are smooth and irregular. Compared with other methods, DSSDNet is better in detecting obvious contour shadows but less effective in detecting sparse shadow distribution regions. The main reason is that the BCE loss function is not better adapted to the type of tiny objects in remote sensing. As shown in Fig. 8(f), the linear target in the red box cannot be detected more accurately. Deeplabv3+ is the same as semantic segmentation methods. A case of wrong detection and omission occurs because of the lack of consideration of the shadow characteristics. The results of DeepLabv3+ are shown in Fig. 8(g), and its detected areas have a gap. For the MTMT method, although it can detect the shadow area better, smoothing occurs for the regular building edges. Compared with other methods, CDANet uses a DB module to add low-level local information to detect shadows. In addition, CDANet replaces the skip connection with CSFC to combine multiscale information. Thus, it can accurately detect irregular distribution shadows. CDANet is better at detecting tiny shadows than other methods. This advantage can be attributed to the hybrid loss function of CDANet, which can flexibly assess the prediction shadow and ground truth.

Two effective comparison experiments referring to the dividing method on the datasets are conducted [1]. The first way is to form the test, evaluation, and test sets according to fixed regions. The second way is to combine the training, test, and evaluation sets together and then redivide them to form the same number of training, evaluation, and test sets. All of our comparison experiments are conducted on the same datasets mentioned above.

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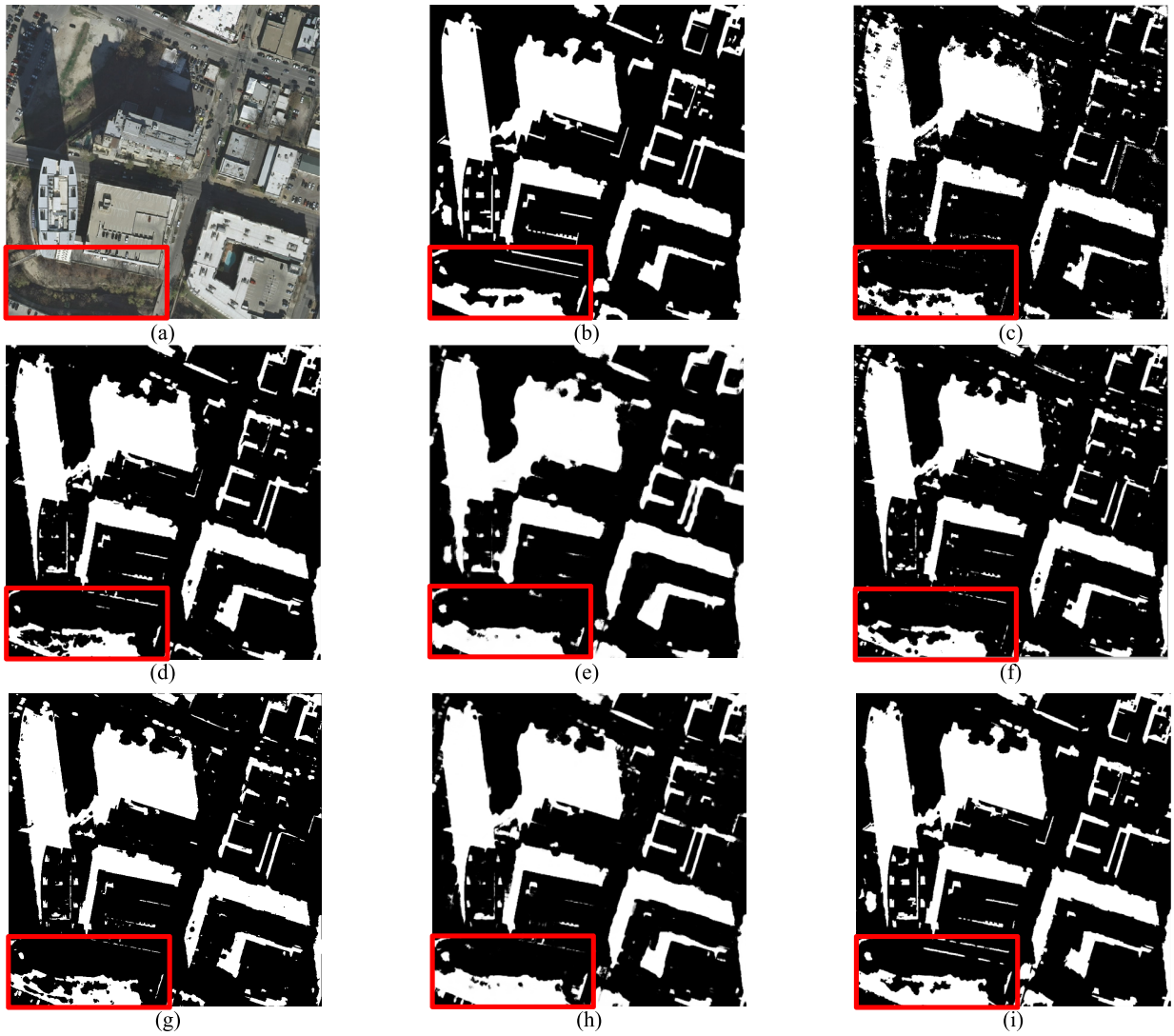


Fig. 8. Shadow detection results using the AISD: (a) images; (b) ground truth; (c) FCN; (d) U-Net; (e) BDRAR; (f) DSSDNet; (g) DeepLabv3+; (h) MTMT; and (i) ours.

TABLE II
TWO DIFFERENT DATASETS COMPOSING BASED ON THE LOCATION

	Training set		Testing set	
	Location	Number	location	number
L1	Austin, Chicago, Tyrol, Innsbruck	435	Vienna	79
L2	Austin, Chicago, Tyrol, Vienna	456	Innsbruck	58

TABLE III
TWO DIFFERENT DATASETS COMPOSING BASED ON THE RANDOMLY EXTRACTING

	Training set	Training patches	Testing set
R1	463	15,899	51
R2	463	15,704	51

The first division method is to test the network effectiveness in detecting shadows formed by different architectural styles. The second division method is to observe the robustness to a randomly distributed shadow. The specific divisions are shown in Tables II and III. The same experimental setup is used for all comparison experiments to ensure fairness.

As shown in Table IV, with the increase in the abundance of surface types, the performance of all methods has fluctuations. However, CDANet shows better robustness on different datasets, and its F1 score is stable at around 92.5% compared with the other networks. Meanwhile, the proposed model is unaffected by black objects and can more accurately separate shadows from black objects than the other networks. Fig. 9 shows some results of CDANet. The F1 scores of the three images are 97.60%, 94.11%, and 93.55%, respectively. Thus, the CDANet result is similar to the ground truth. As shown in Fig. 9(c), CDAnet can effectively distinguish between black objects and shadow areas, and does not detect black cars

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TABLE IV
AVERAGE F-SCORE FOR DIFFERENT METHODS BASED ON LOCATION-BASED AND RANDOM-BASED DATASETS

Datasets	FCN	U-Net	BDRAR	DSSDNet	DeepLabv3+	MTMT	CDANet
L1	86.29%	88.59%	84.68%	90.95%	91.51%	89.84%	92.29%
L2	86.47%	87.73%	85.41%	90.84%	90.79%	90.66%	92.02%
R1	85.31%	86.60%	88.14%	89.42%	90.80%	90.51%	93.12%
R2	85.08%	84.79%	87.89%	88.68%	90.08%	90.30%	91.02%

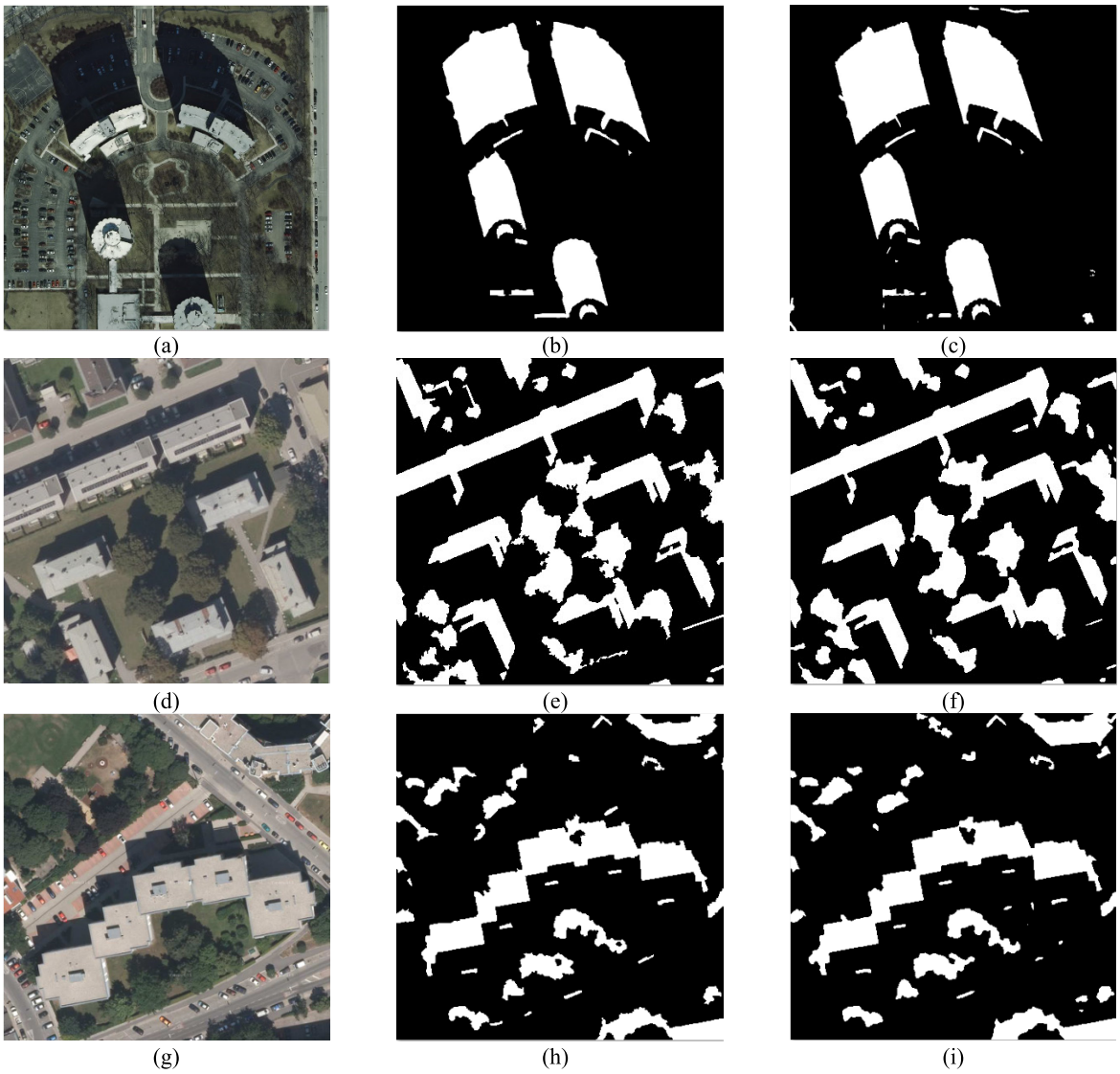


Fig. 9. Shadow detection on three HSR images with the proposed CDANet: (a)–(c) Image #1, ground truth, and shadow detection result, respectively; (d)–(f) Image #2, ground truth, and shadow detection result, respectively; and (g)–(i) Image #3, ground truth, and shadow detection result, respectively.

as shadows. As shown in Fig. 9(f), CDAnet can also detect irregular tree shadows very well, and its detection result is very similar to the ground truth.

E. Experiment 2: SSAD

The SSAD has only 103 500 × 500 infrared-red and green image synthesis images. According to the principle of category balance, 79 images are selected and then cropped to form 1900 patches of 256 × 256 patches for training, and three images are selected and then cropped to form 60 patches of 256 × 256 patches for validation. The rest of the 21 images are used as a test set to evaluate the prediction effect of the model. In SSAD, more shadows are formed by trees and woodlands, and urban shadows are dense. The color of the water in this dataset is very close to the shadows, which also complicates shadow detection. As shown in Table V, our proposed network has a lower error rate on the SSAD. Thanks to the hybrid

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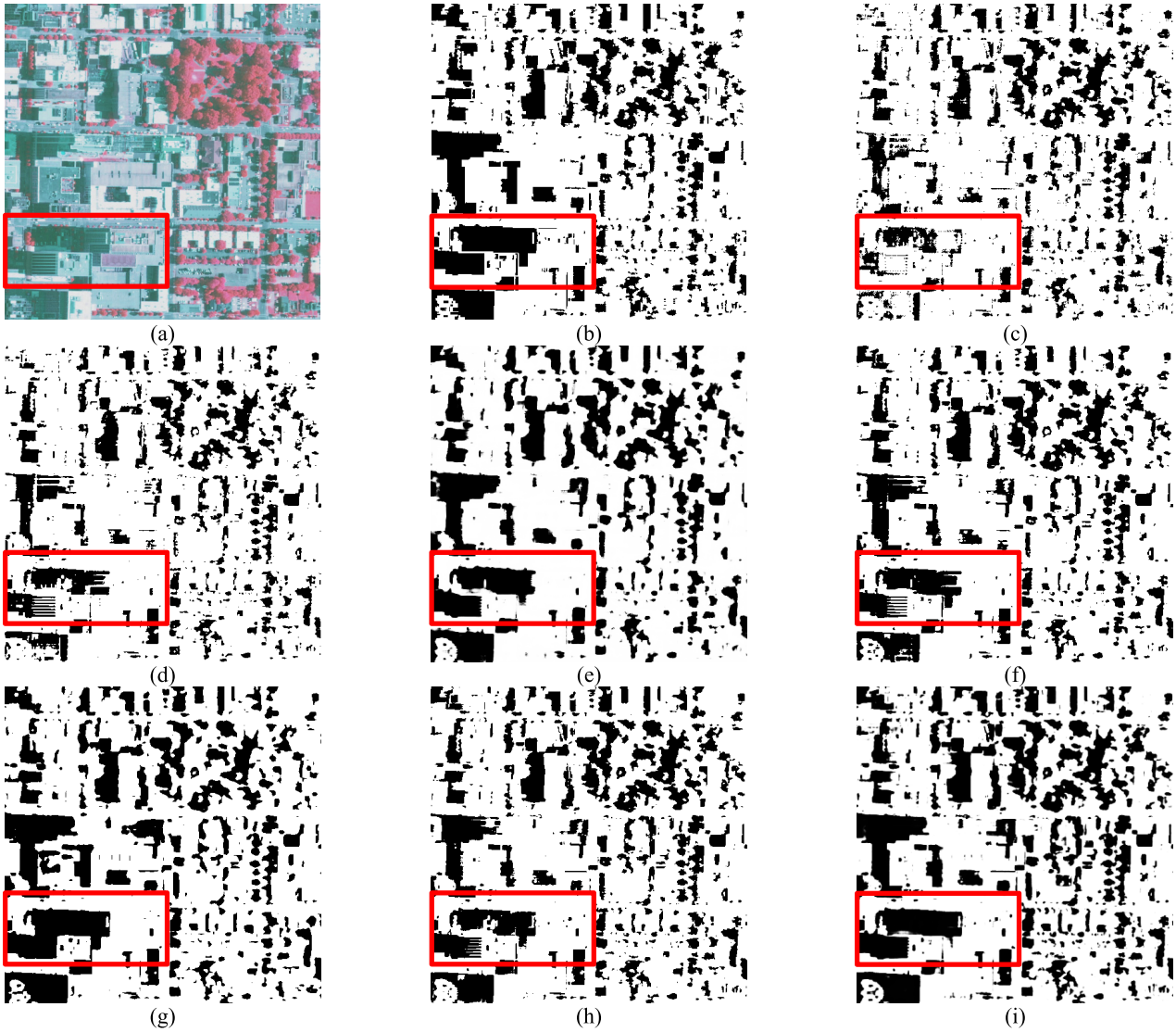


Fig. 10. Shadow detection results using the SSAD: (a) images; (b) ground truth; (c) FCN; (d) U-Net; (e) BDRAR; (f) DSSDNet; (g) Deeplabv3+; (h) MTMT; and (i) proposed method.

TABLE V
COMPARISON OF DIFFERENT SHADOW DETECTION METHODS ON THE SSAD

Model Name	F1(%)
FCN	75.05±1.14
U-Net	77.68±0.50
BDRAR	74.76±0.98
DSSDNet	79.31±0.93
DeepLabv3+	76.64±0.52
MTMT	79.02±0.35
Ours	**83.96±0.23**

loss function, CDANet has better accuracy in irregular shadow detection than the other networks. In addition, the other networks cannot detect such dense shadows well, and certain shadow regions will always be missing. The proposed network can better adapt to this situation, and the fragmented shadow detection is excellent. The visualization results are shown in Fig. 10.

F. Ablation Study

Ablation experiments with different settings are conducted to verify the influence of each component. UNet is used as a baseline for a fair comparison, and it is evaluated using the AISD. The results are enhanced by 2.49% after the DB block. This result demonstrates the effectiveness of the DB block. As shown in Table VI, when the CSFC is attached, the results improve from 89.98% to 91.96%. Thus, the CSFC module enhances the occurrence of insufficient information integration. Finally, the combination of the hybrid loss and previous components achieves the best performance. The ablation study shows that each component of the proposed model is necessary to obtain the best results.

G. Sensitive Analysis for Shadow Detection

The coefficient λ is essential for constructing a robust hybrid loss function, and its effect on the shadow detection

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TABLE VI

ABLATION STUDY WITH DIFFERENT COMPONENTS COMBINATIONS ON AISD

Baseline	DB	CSFC	Hybrid loss	F1
√				87.51%
√	√			89.98%
√	√	√		91.96%
√	√	√	√	92.99%

TABLE VII

SENSITIVITY ANALYSIS OF λ ON DIFFERENT DATASETS

λ	F1 (%)	
	AISD	SSAD
0	91.96%	80.25%
0.2	92.49%	**83.96%**
0.5	**92.99%**	83.00%
0.8	92.41%	83.04%
1	92.23%	80.34%

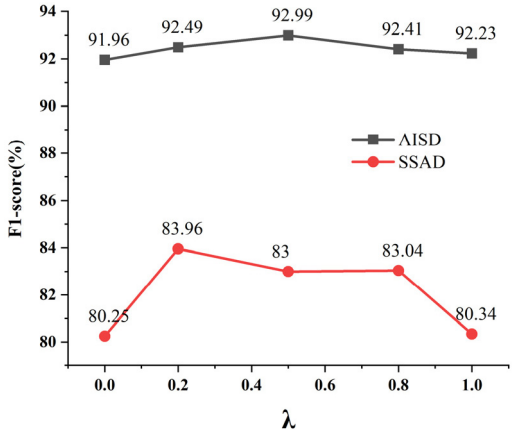


Fig. 11. Sensitivity analysis of λ in two datasets, respectively.

results is studied. We set values of 0, 0.2, 0.5, 0.8, and 1.0 to verify the effect of different Jaccard-hinge loss and BCE loss weights. As shown in Table VII and Fig. 11, with the increase in the parameter λ, the F1-score in the two datasets increases and then decreases. This phenomenon illustrates that the weights of the BCE function and Jaccard-hinge function in the loss function significantly affect the detection results. In addition, the effect of different λ values is more significant for the SSAD than the AISD. This result can be ascribed to the more irregular and small shadows in the SSAD than in the AISD, demonstrating the importance of the Jaccard-hinge function for the extraction of tiny shadows. When the weight of the Jaccard-hinge loss function is too large, it will affect the efficiency of the BCE function and decrease the detection accuracy because the regular shading becomes rounded. By contrast, when the Jaccard-hinge loss takes up a small proportion, the probability of detecting regular shadows increases, and the tiny local shadows are lost. As shown in Table VII, the

TABLE VIII

RUNTIME AND PARAMETER COMPARISON WITH OTHER METHODS

Method	Time(s)	Param(M)
FCN	0.045	71.15
U-Net	0.128	65.95
Deeplabv3+	0.052	453.48
BDRAR	0.176	164.01
MTMT	0.145	169.21
DSSDNet	0.111	42.43
CDANet	0.171	164.23

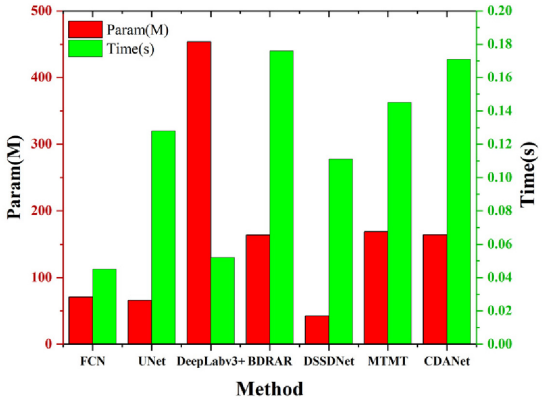


Fig. 12. Running time and algorithm complexity about different models of AISD.

accuracy of the hybrid loss function increases on AISD and SSAD by 1.03% and 3.71%, respectively, compared with the traditional BCE loss function. The best accuracy of 92.99% is achieved on the AISD when λ is 0.5. SSAD has more fragmented shadows, such as trees, and the improvement in accuracy is more obvious. The best accuracy of 83.96% is achieved on the SSAD when λ is 0.2.

All experiments are conducted with an NVIDIA GeForce GTX 2080 Ti GPU to explore the running time and algorithm complexity of various methods. A batch size of 10 is used in every experiment. The size of the input image is 256 × 256. As shown in Table VIII and Fig. 12, our model has 164.23M parameters, and the single batch running time of the model is 0.171 s. Although the CDANet framework has higher runtime and algorithm complexity than FCN, U-Net, and DSSDNet, its accuracy, and visualization results are much better than these methods, as shown in Figs. 8 and 10. In addition, compared with state-of-the-art shadow detection methods such as BDRAR and MTMT, the CDANet framework achieves higher accuracy with fewer parameters and less time complexity. Furthermore, CDANet has less complexity and higher accuracy than DeepLabv3+.

H. Spectral Variability Analysis

As known to all, spectral variability is a universal phenomenon due to illumination conditions and topography, atmospheric effects, or even the intrinsic variability of the material, which bring big challenges for HSR image

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Original image Simulated image

Fig. 13. Comparison of the original and processed images in the test set.

TABLE IX
COMPARISON OF DIFFERENT SHADOW DETECTION METHODS ON SPECTRAL VARIABILITY SIMULATED BY AISD

Model Name	F1(%)
FCN	79.65
U-Net	87.56
BDRAR	87.26
DSSDNet	87.18
DeepLabv3+	77.90
MTMT	89.29
Ours	**90.50**

interpretation [79]. Shadows appear when the sunlight is completely or partially obscured by an object, all images quality will be affected due to variation in illumination. To explore the effect of spectral variability on shadow detection, we analyzed the shadow detection results of the original image and corresponding simulated image, which is obtained by modifying the spectral tones of the original image. As shown in Fig. 13, to simulate the effect of illumination variation, we divide the test set from the AISD with 51 HSR images into three randomly averaged sets, and make random adjustments to the illumination tones of the HSR images.

We follow the same operation of Section IV-C to train on the original train set and test on the simulated images from the AISD. As shown in Table IX, with the illumination variation of the image, features such as grasses are more similar to shadows, which results in significant performance degradation for all networks. However, the CDANet shows better performance on simulated images, and its F1-score is stable at around 90.5% compared with the other networks. The F1-score of CDANet is more stable and accurate compared to other models.

V. CONCLUSION

This article proposes a context saliency detection network based on hybrid loss function (CDANet) for HSR imagery shadow detection. In CDANet, a DB module is used to alleviate the information loss problem, and then the information in the bridge is connected with the RD module to retain the effective shadow features. The decoder uses a connection operation with fused contextual information to enhance the global spatial relationships. Finally, a novel loss function is adopted to discover the distribution of more scattered shadows. Experiments performed on two shadow datasets (AISD and SSAD) demonstrated that the proposed shadow detection method could more effectively detect small object shadows than other methods. Moreover, our proposed model can maintain better detection on images with chromatic aberration. In our future work, we plan to use the shadow edge as auxiliary information to improve the accuracy of shadow detection. Besides, the model runtime and algorithm complexity should be reduced to meet the standards of a lightweight network. We plan to convert RGB images into Lab images for feature learning due to the shadows being more sensitive to the L channel.

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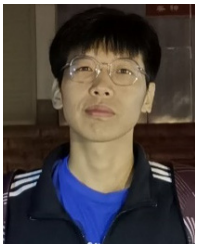
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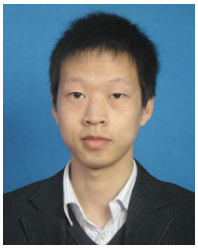
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